Hayat Karabulut

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Profile

MSc student in Artificial Intelligence and Robotics at Sapienza University of Rome, with a BSc in Applied Computer Science and AI (110/110 with honors). Experience in machine learning, deep learning and vision-language data pipelines, plus software development, UX design and leading technical teams.

Education

Sapienza University of Rome — MSc, Artificial Intelligence and Robotics · Oct 2026 – present

Sapienza University of Rome — BSc, Applied Computer Science and Artificial Intelligence · Sep 2023 – Oct 2026

- Graduated 110/110 with honors.
- Thesis: built a data preprocessing pipeline that converts vision-language datasets with noisy, human-annotated captions into supervision for hyperbolic embeddings of images and text descriptors.

The Koc School, Istanbul — International Baccalaureate Diploma · Sep 2018 – Jun 2023

Technical Skills

- **AI and machine learning:** machine learning, deep learning, vision-language data, embeddings, dataset preprocessing and cleaning
- **Programming:** Python, Java, C++, JavaScript, Go, Dart, SQL
- **Web and software:** HTML, CSS, React, Vue.js, web and software architecture, object-oriented programming
- **Design:** Figma, Procreate, Adobe Illustrator
- **Languages:** Turkish (native), English (C1), Italian (A2), French (A1)

Relevant Coursework

- **AI and machine learning:** Artificial Intelligence, Machine Learning, Advanced Machine Learning, Deep Learning

- **Software:** Python Programming, Java Programming, Object-Oriented Programming, Web and Software Architecture, Robot Programming in C++ (in progress)
- **Robotics and systems:** Robotics, Computer Architecture
- **Data and optimization:** Data Management and Analysis, Linear Optimization, Signal Processing, Bioinformatics
- **Business:** Business and Computer Science (designed business models)

Experience

Project Team Manager — FutureTea (remote startup) · Jan – Jul 2025

- Managed the project team building an AI app for checking the reliability of sources.
- Designed the full UX of a separate mental health app in Figma, Procreate and Adobe Illustrator.

Head of IT and Deputy Secretary-General of Publications — MUNDP (Model UN Development Conference) · 2019 – 2022

- Updated the conference website and built a demo database for it.
- Designed the sponsorship document, invitation package and handbook.
- Handled live issues during the event, such as a website crash.

Tutor and President — Coding Club, The Koc School · 2019 – 2023

- Built a one-year curriculum and taught Python and Pygame to middle and high school students.

Work placement — WeWalk · Jul 2020

- Researched competitors to inform the company's landing and campaign page, and presented findings with a Figma mockup.

Selected Projects and Competitions

Thesis: Hyperbolic embeddings from noisy vision-language data · Sapienza, 2026

- Designed a preprocessing pipeline that turns noisy human-written captions into usable supervision for hyperbolic image–text embeddings.

EvZane: app for bedridden patients' medical supply needs · TEKNOFEST, 2021

- Competed in the social innovation category and reached the 3rd stage.
- Designed the front-end mockup of the mobile app and wrote the solution plan report.

Volunteering map web platform · SAGE Global Entrepreneurship, 2021 – 2022

- Helped develop the website for a map of places needing volunteers or donations.
- Team won an award for a business model supporting homemaker women in Besiktas, Istanbul.

Video game design and fundraising · TSA (Technology Student Association), 2019 – 2023

- Video Game Design team member for 2 years; President for 1 year.
- Wrote a sponsorship package and raised $5,000 to fund the team's travel.

FIRST Robotics Competition · 2018 – 2023

- Head of the coding division; won the Rookie Award.
- Created a one-year curriculum to prepare new members for competition.

Awards and Additional

- **Awards:** First Robotics Competition Rookie Award; ECIS International Understanding Award (2022); LP Computer Education Award and LP Science Award; Academic High Honors, grades 9–11
- **Volunteering:** mentor at YGA (Foundation for raising leaders and entrepreneurs), Mar 2020 – present; ran science activities for children and helped raise donations for science kits
- **Research:** wrote a paper on lab-on-a-chip technology and how it varies with the organ being simulated (grades 11–12)